

CASE 02

From manual map edits to a real-time event platform

One publishing path for public traffic information and commercial content in navigation

Lead Product Manager · end to end platform owner

Anonymised case study · 2020 to 2021



Illustrative image, not the product.

GOAL

Publish short-lived traffic events and commercial content into navigation, fast and precisely

USERS

Traffic operations, marketing, advertising clients and drivers

MY ROLE

Platform design, requirement decomposition and delivery as the single product owner

SCALE

Seven teams, 30+ engineers, six months to full release

OUTCOME

Seconds to activation, 10+ clients, RMB 6M+ recognised revenue

Two unrelated demands, one publishing bottleneck

PUBLIC INFORMATION

Short-lived traffic events

Closures, city or area restrictions and temporary controls that lose their value within hours.

MONETISATION

A free product needed revenue

Clients wanted to reach drivers before, during and after a drive, by voice or on the map.

THE SAME FIVE QUESTIONS

- Who receives it
- Where it triggers
- What users see or hear
- How often it appears
- When it starts and ends

ONE PATH TO BUILD

Different business purposes, identical product mechanics. Building twice would have meant maintaining two half-solutions.

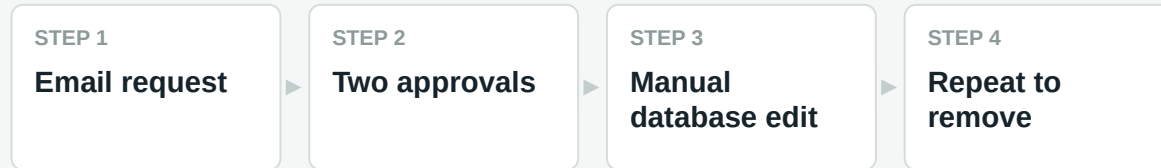
Governance, not code, would carry the difference.

Neither demand could be served by the publishing flow that existed.

Publishing an event meant an email request, two approvals and a data team editing the base map by hand. Removal meant repeating all of it. Typical turnaround was one to two days.

Speed was the symptom, expressiveness was the limit

THE VISIBLE PROBLEM



1 to 2 days
typical turnaround

THREE WAYS IT FAILED

An event could expire before it was published

Hand-offs introduced interpretation errors

A manual edit could attach content to the wrong road

WHAT THE OLD DATA MODEL COULD EXPRESS

one whole road, all users, no end time

Nothing narrower: not a segment, an area, a route, or an audience.

WHAT THE BUSINESS NEEDED TO EXPRESS

a road segment

a drawn area

a planned route

a POI

an audience

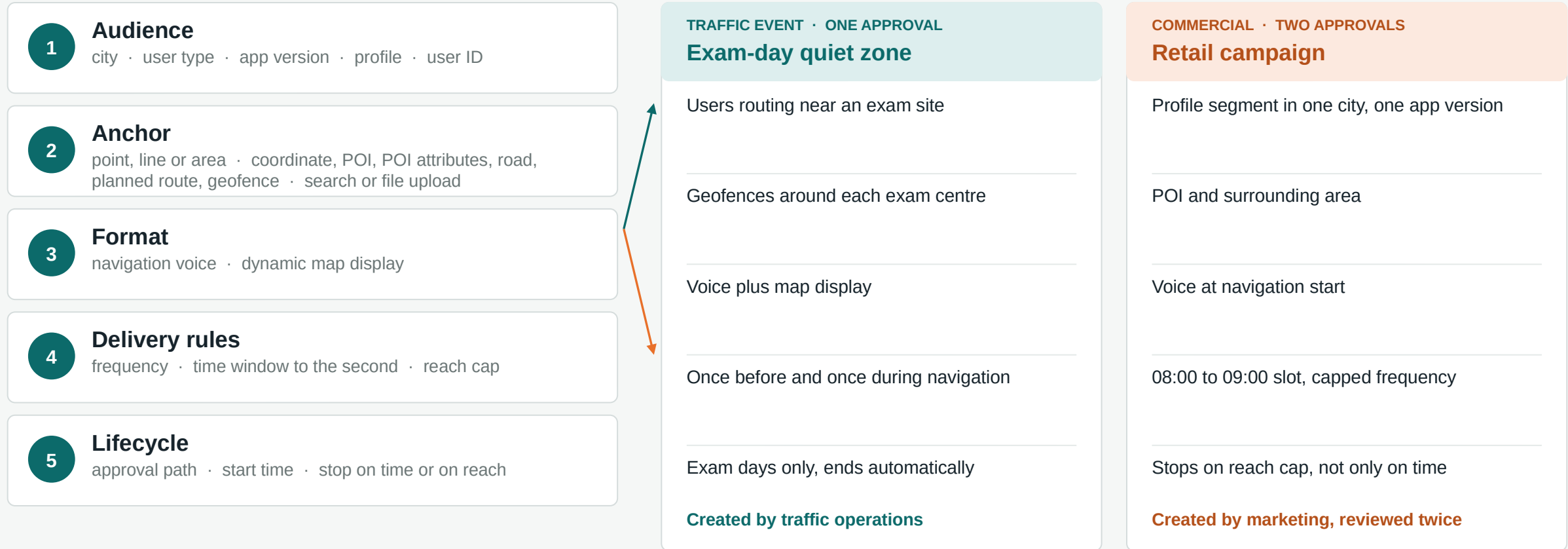
a time window

Automating the workflow alone would have produced the same coarse events, only faster.

So the project was not a workflow tool. It was a new way to describe an event.

Five dimensions, recombined instead of rebuilt

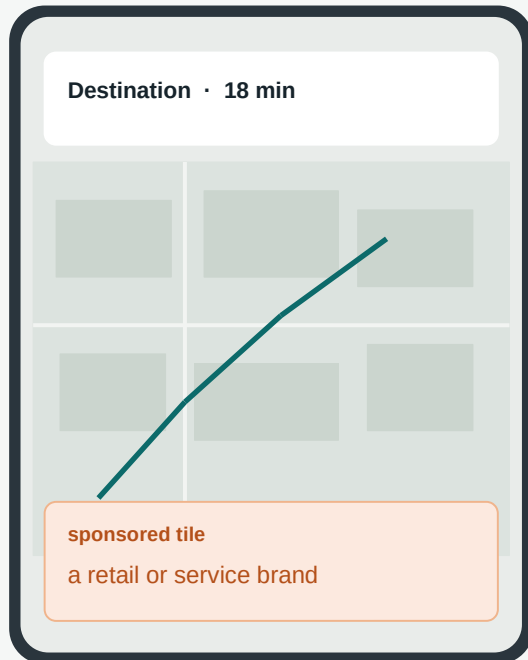
THE FIVE DIMENSIONS



Same five dimensions produced a public-safety notice and a paid campaign. Only the approval path and the delivery rules differed.

Before the platform, an event could attach to an entire road and nothing narrower. Each new request meant another manual database edit.

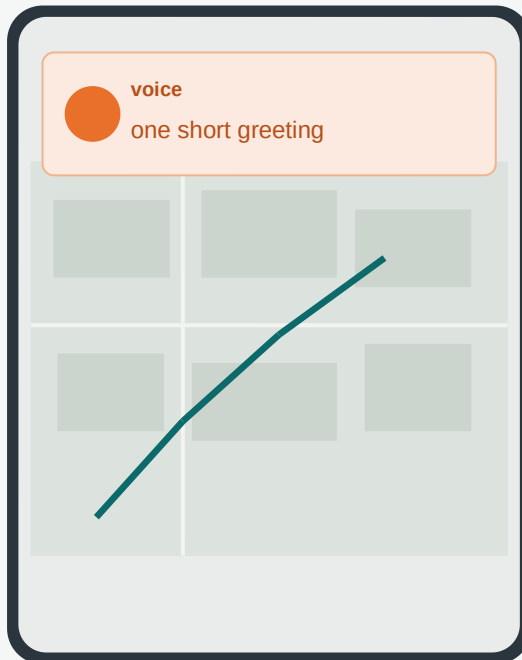
Four moments in a drive, matched to how much they interrupt



before the drive

Route planning

map display · a retail or service brand

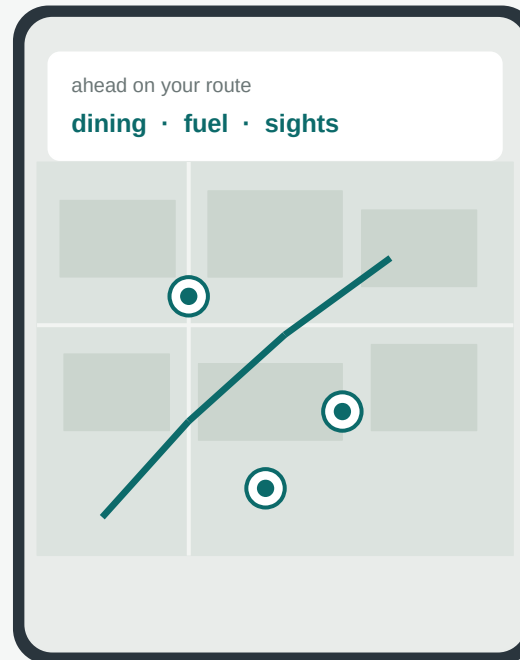


first seconds

Navigation start

voice · an energy drink category

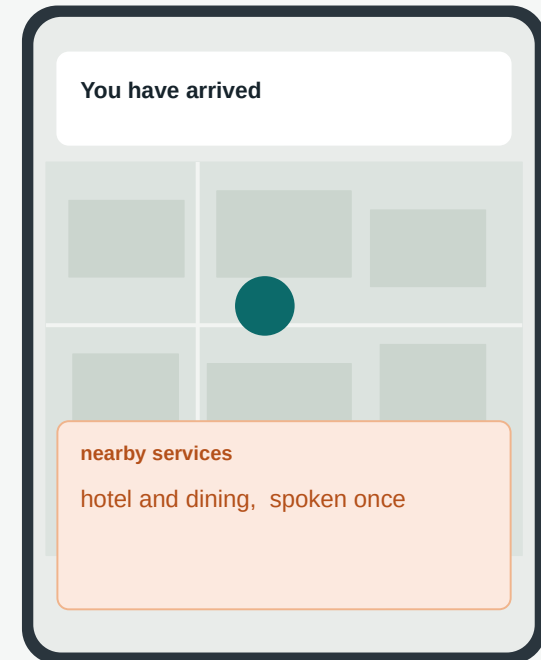
highest contention



idle stretches

En route

voice and map · dining, attractions, fuel



drive complete

After arrival

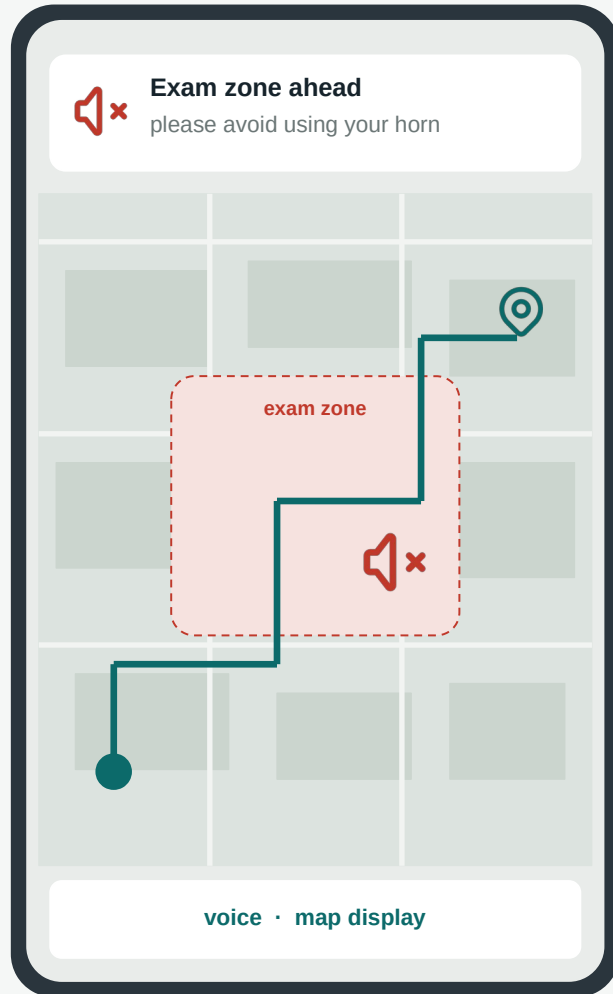
voice · hotel and dining services

highest contention

En route content is location-relevant and tolerant of length. Start and arrival are interruptions, so they stay short — and ten or more clients wanted exactly those two slots.

Illustrations drawn for this portfolio. Clients are shown as industry categories; no names or campaign content are disclosed.

One event, drawn once and delivered to the right drivers



HOW THE EVENT WAS CONFIGURED

- 1 Audience**
Drivers whose route crosses the zone
- 2 Anchor**
A geofence drawn around each exam centre
- 3 Format**
Voice before the drive and on approach
- 4 Delivery rules**
Once each, no repeat within the zone
- 5 Lifecycle**
Exam days only, ends automatically

BEFORE THE PLATFORM

- Email request, two approvals, manual database edit
- One to two days, often after the event mattered
- Attachable to a whole road, not to a drawn area
- Removal needed a second request

Now: configured in one form, live seconds after approval

Illustration. Drawn for this portfolio, not a screenshot.

A public-safety notice and a paid campaign were the same kind of object. Only the approval path and the rules changed.

One form, one approval trail, one live dashboard

Draft
operations

Submitted

Approval
named approver

Live
auto activation

Ended
time or reach cap

EVENT FORM

1 Timing

start

immediately after approval

time window

08:00:00 – 09:00:00

stop condition

time · reach cap

approval path

commercial: two steps

2 Audience and frequency

city and user type

commuters, one city

profile segment

retail interest

app version

5.x and above

frequency cap

1 voice per user

3 Content and anchoring

voice

text and trigger moment

anchor type

point · line · area

map display

position, style, timing

anchor input

search or file upload

LIVE DASHBOARD

time

event

area

audience

version

reached
users reached

plays
voice plays

displays
map displays

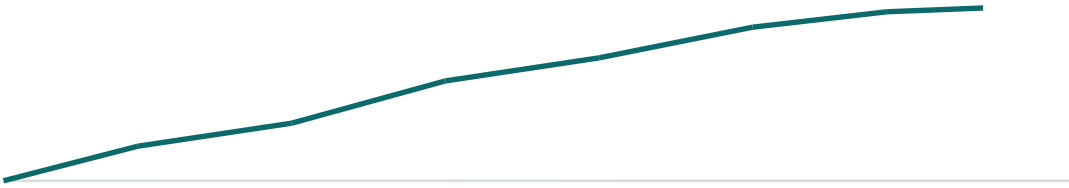
avg
plays per user

avg
displays per user

max
worst-case exposure

REACH AGAINST THE CAP

reach cap · campaign stops here



Operators could stop on a schedule or on volume. Frequency and per-user exposure were first-class metrics, not afterthoughts.

Redrawn wireframe. No internal screens, client names or field values are shown.

The real question was how far to abstract

THREE WAYS TO BUILD IT

Build per client

Fast for the first request. Every change means another manual database edit.

CHOSEN

Split into composable dimensions

Only works if the right dimensions were chosen.
Guess wrong and the work is wasted.

A fully generic rule engine

Maximum flexibility that operations could not use and delivery could not carry.

HOW I CHOSE THE DIMENSIONS

1

Look backwards

Which dimensions past requests had actually varied on: audience, location, frequency, lifecycle.

2

Look forwards

Which levers a commercial campaign would plausibly need before any client asked for them.

DELIBERATELY NOT BUILT

A priority and collision engine

High-priority events could have pre-empted low-priority ones automatically. The more complex the rules, the less a client could predict whether their campaign would actually run.

Instead

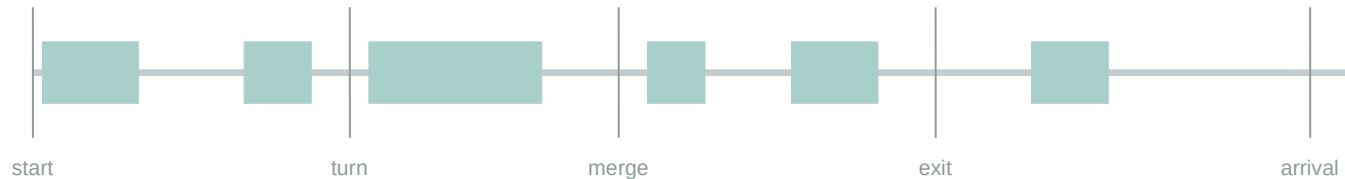
Operations separated campaigns by time slot and by area. Manual, but predictable for the client.

In early monetisation a client needs predictability more than optimal allocation. Sophistication was the risk, not the goal.

One dimension the model treated as a constant

VOICE WINDOWS ALONG ONE ROUTE

available gaps between navigation instructions



a long commercial message needs this much

WHAT HAPPENED

CLIENT REQUEST

A longer message in the idle stretch of a drive

WHAT WE HIT

Many routes had no gap long enough to play it

ROOT CAUSE

Format was modelled as a choice, not as capacity

HOW WE RESOLVED IT

Shorter copy, moved to the pre- and post-navigation touchpoints, with stronger map display to carry the message.

No new feature was built. The fix came from recombining dimensions the model already had.

WHAT I WOULD DESIGN DIFFERENTLY

• Treat broadcast capacity as a dimension

Model the space a route can carry, not only the format an operator picks.

• Run a real test pass before approval

Publish to a designated test audience first, so operations sees the real result instead of imagining it.

The model absorbed the problem, but it should have predicted it. Capacity was the dimension I left as a constant.

Both changes are how I would design it today, not capabilities that shipped at the time.

What the platform proved, and what it did not

DIRECTLY OBSERVED

Seconds

to activation once an approver clicks through, against one to two days of manual coordination before

10+

advertising clients live in the same period

17+

commercial placements per month, plus about 5 traffic events

Two business lines, one publishing path. Governance differed: traffic events took one approval, commercial content took two.

COMMERCIAL RESULT

RMB 6M+

recognised revenue in the six months following full release

How to read this

Clients bought the ability to publish into navigation moments. The platform itself was never sold externally.

Revenue was received and recognised, not booked or forecast.

Roughly ten clients over six months makes the per-client scale ordinary, not exceptional.

Delivered inside a navigation product with over 100 million daily active users. That is the environment, not the reach of any single event.

NOT PROVEN

Delivery, not conversion

The platform counted users reached, voice plays and map displays, with per-user and peak exposure alongside them.

It had no click, footfall or conversion funnel, so it could show that a message was delivered but not that it changed behaviour.

Next steps I would take

- Treat advertising capacity as a product capability, with priority and conflict rules
- Add a conversion funnel so effect is judged on more than plays and displays

Public use

On national exam days the platform published quiet-zone notices around every exam centre, reaching drivers who had routed nearby before and during navigation.